

TASTE-Rob: Advancing Video Generation of Task-Oriented Hand-Object Interaction for Generalizable Robotic Manipulation

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<https://taste-rob.github.io>

Abstract

We address key limitations in existing datasets and models for task-oriented hand-object interaction video generation, a critical approach of generating video demonstrations for robotic imitation learning. Current datasets, such as Ego4D [16], often suffer from inconsistent view perspectives and misaligned interactions, leading to reduced video quality and limiting their applicability for precise imitation learning tasks. Towards this end, we introduce TASTE-Rob — a pioneering large-scale dataset of 100,856 ego-centric hand-object interaction videos. Each video is meticulously aligned with language instructions and recorded from a consistent camera viewpoint to ensure interaction clarity. By fine-tuning a Video Diffusion Model (VDM) on TASTE-Rob, we achieve realistic object interactions, though we observed occasional inconsistencies in hand grasping postures. To enhance realism, we introduce a three-stage pose-refinement pipeline that improves hand posture accuracy in generated videos. Our curated dataset, coupled with the specialized pose-refinement framework, provides notable performance gains in generating high-quality, task-oriented hand-object interaction videos, resulting in achieving superior generalizable robotic manipulation. To foster further advancements in the field, TASTE-Rob dataset and source code will be made publicly available on our website <https://taste-rob.github.io>.

1. Introduction

Robotic manipulation plays an essential role in daily life, assisting with tasks like object handling, liquid pouring, surface cleaning, and drawer operation. Powered by the success of Imitation Learning (IL) [8–10, 15, 20, 22, 42, 48,

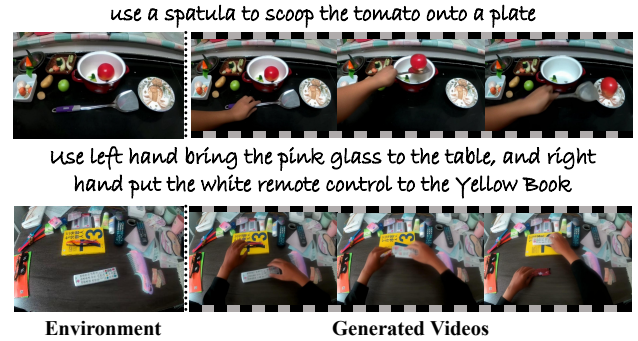


Figure 1. **TASTE-Rob** generates high-quality task-oriented hand-object interaction videos, enabling robust generalization in robotic manipulation.

[56, 57, 64], current robots can imitate actions from video demonstrations but require a nearly identical environment in the recorded video, limiting their generalization to novel scenes where factors like the position or movement direction of the manipulated object differ.

To solve the issue of limited generalization, recent research combines IL with generative models [2, 3, 5–7, 12, 25, 47, 58], like video diffusion models (VDMs), to create adaptable demonstrations and enable more generalizable robotic manipulation [12, 25, 47]. Specifically, these works develop VDMs to generate robot-object interaction videos as demonstrations for IL. While robot-object interaction VDMs enhance generalization performance, they are constrained by limited robot datasets [4, 8, 21, 24, 51]. A scalable alternative is generating hand-object interaction (HOI) videos, which leverage extensive human manipulation video datasets. Although hand-object interaction differs from robotic motion, advanced policy models [1, 5, 9, 22, 48, 57, 65] can bridge this gap effectively, achieving remarkable performance through IL. However, even if existing powerful VDMs are able to generate remarkable human videos, as shown in Figure 7, they struggle with generating high-quality, task-oriented hand-object interaction videos with accurate task understanding.

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To enhance video generation quality, Ego4D [16], a large-scale data containing 83,647 ego-centric hand-object interaction videos, has proven useful in robotic manipulation tasks [23, 32, 53]. However, Ego4D presents two notable limitations for our purpose, as shown in Appendix 8. Firstly, the camera perspective in each clip is inconsistent, whereas video demonstrations for IL are typically recorded from a fixed viewpoint. Secondly, since these clips are extracted from longer videos that encompass multiple tasks, some clips contain partial interactions that overlap with others, resulting in misalignment between video clips and language instructions. Both limitations negatively impact the quality of video generation.

To address these above limitations, we introduce TASTE-Rob, a pioneering large-scale dataset designed to enhance the generation quality of task-oriented hand-object interaction videos. TASTE-Rob contains 100,856 ego-centric videos with well-aligned language instructions. Unlike Ego4D, TASTE-Rob is crafted to ensure clarity and consistency – each video captures a single, complete interaction corresponding to its instruction and is recorded independently with a fixed camera viewpoint, mirroring the conditions needed for reliable imitation learning demonstrations. By resolving these key issues, TASTE-Rob provides a robust foundation for advancing research in video generation, imitation learning, and related fields.

After fine-tuning a VDM on our curated dataset, we achieve high-quality video generation of task-oriented hand-object interactions. While the model demonstrates proficiency in identifying target objects and determining appropriate coarse interactions, it exhibits inconsistencies in hand poses, with the grip changing unnaturally during manipulation. This limitation presents a critical challenge for robotic imitation learning where precise and stable hand-object interactions are essential. To address this issue, we further propose a dedicated three-stage pipeline for pose refinement. First, we generate a coarse manipulation video conditioned solely on language instructions and environment images. Second, we refine the hand pose sequences using a Motion Diffusion Model (MDM) [45], ensuring realistic grip poses. Finally, we regenerate the videos by incorporating the revised pose sequences as an additional conditioning factor. This pipeline enhances grasping realism in generated videos, enabling robots to execute novel tasks with greater accuracy and adaptability in unseen scenarios. Extensive experiments demonstrate that our proposed TASTE-Rob dataset, combined with the pose-refinement pipeline, achieves significant performance gains, surpassing state-of-the-art methods in the quality of generated hand-object interaction videos and achieving accurate robotic manipulation.

Our contributions can be summarized as follows:

- We introduce TASTE-Rob, a large-scale dataset contain-

ing 100,856 ego-centric hand-object interaction videos, each specially tailored for training high-quality video generation model for robot imitation learning.

- We develop a three-stage pose-refinement pipeline that significantly enhances the fidelity of the hand posture in the generated videos.
- We set a new state of the art in generating task-oriented hand-object interaction videos, which further boosts the performance of robot manipulation.

2. Related Works

Imitation Learning from video demonstrations. IL enables robots to learn from expert [8, 10, 15, 42, 64] or human [9, 20, 22, 48, 56, 57] video demonstrations by training policies to directly replicate expert actions through supervised learning. While these methods require video demonstrations from environments identical to the robot execution setup, they exhibit limited generalization to novel scenes and tasks. Alternatively, existing works address this challenge by generating demonstrations, such as videos [5, 23, 32, 53], optical flows [52, 58], and images [7], successfully achieving better generalization. Among existing approaches, video demonstration generation represents the most straightforward solution. Yet, robot manipulation video generation [23, 32, 53] is constrained by limited robot datasets, while human hand video generation [5] is hindered by the absence of effective HOI video generative models. In this work, we explore enhancing HOI video generation quality to facilitate robot manipulation.

Ego-Centric HOI Video datasets. To enhance video quality, we propose a specialized HOI video generation model, which requires a large-scale, high-quality ego-centric dataset. Existing ego-centric HOI datasets have various limitations: insufficient resolution [27, 28, 30, 43], limited scale [13, 27, 28, 30, 35, 43], and restricted scene diversity [11, 13, 28]. While Ego4D [16] provides diverse, large-scale HOI videos, its varied camera perspectives and imprecise action-language alignment pose challenges for IL video demonstrations. Therefore, we introduce TASTE-Rob, containing 100,856 videos specifically designed for our task.

Diffusion Models for Video Generation. Based on the success of VDMs [18], researchers have achieved high-quality video generation with various conditioning inputs: text [33, 41, 61, 62] and images [26, 38, 54, 55, 60]. Image-to-video generation aligns well with our task. While these models excel at producing realistic colors and temporal coherence, they struggle with significant motion variations. Motion diffusion models [19, 36, 45, 50, 59] address this limitation by leveraging high-quality human body models [39] to generate motion control parameters. In this work,

Orientation($^{\circ}$)	0-90	90-180	180-270	270-360
Proportion(%)	33.82	47.7	9.59	8.89

Table 1. **Palm orientation distribution of grasping poses.** We analyze the palm orientation distribution of grasping poses across TASTE-Rob. Specifically, the degree of orientation denotes the angle between the palm normal direction and the positive x -axis of the frame plane.

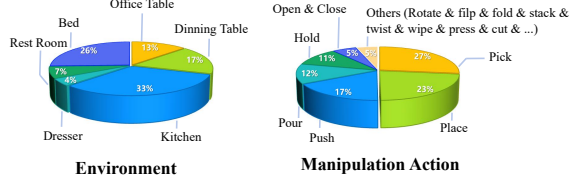


Figure 2. **Distribution of environment and HOI task.** In TASTE-Rob, we collect ego-centric task-oriented HOI videos with diverse tasks under diverse environments.

we combine the advantages of VDM and MDM through our pose-refinement pipeline to generate HOI videos with realistic grasping poses.

3. Building TASTE-Rob Dataset

We collect a massive and diverse ego-centric task-oriented HOI video dataset **TASTE-Rob**, which includes 100,856 pairs of video and their corresponding language task instruction. To serve for HOI video generation, TASTE-Rob is required to achieve following goals: 1) Each video is recorded with a static camera perspective and contains a single action that closely aligns with the task instruction. 2) It encompasses diversified environments and tasks. 3) It features a variety of hand poses across different HOI scenarios.

3.1. Collection Strategy and Camera Setting

To achieve the first goal, we utilize multiple cameras equipped with a wide-angle lens capable of capturing 1080p ego-centric videos. We make the following improvements during each recording process.

Firstly, since our data collection aims to generate task-oriented HOI videos for demonstrations in IL, where demonstrations are typically recorded from fixed camera viewpoints for effective robot imitation learning, we ensure that no camera perspective variation occurs during recording. In addition, as shown in Figure 12, we align the camera perspective specifically to match the head-mounted camera setup of Ego4D [16], ensuring consistency with the ego-centric perspective.

Our second objective is to ensure precise alignment between language task instructions and video actions in TASTE-Rob, a crucial aspect for maintaining the action integrity in generated HOI videos. Unlike Ego4D [16], which captures extended recordings of daily activities via head-mounted cameras and later segments them into shorter clips,

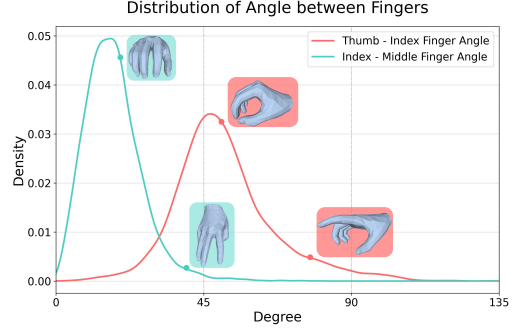


Figure 3. **Inter-finger angle distribution of human hands.** We analyze the distribution of inter-finger angles, specifically between the thumb-index and index-middle finger pairs, across TASTE-Rob. Specifically, higher degrees mean higher inter-finger angles.

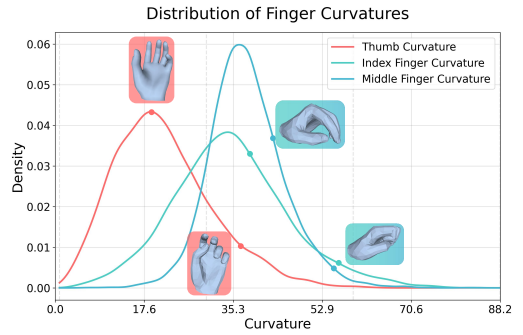


Figure 4. **Finger curvature distribution of human hands.** We analyze the distribution of finger curvatures, specifically that of thumb, index finger and middle finger, across TASTE-Rob. Specifically, higher degrees mean higher curvature.

we employ a more controlled collection protocol: 1) Each video is strictly limited to under eight seconds in duration and captures a single action. 2) Collectors follow a structured recording process: they press the “start recording” button, execute the specified HOI task according to the provided instruction, and stop recording upon task completion. This approach ensures a precise correspondence between actions and task instructions.

3.2. Data Diversity

Distribution of environment and task. To promote broad generalization, videos in TASTE-Rob are recorded across diverse environments and cover a wide range of HOI tasks. As shown in Figure 2, environments include locations such as kitchens, bedrooms, dining tables, office tables, etc. Collectors are required to interact with various commonly used objects, and perform tasks including picking, placing, pushing, pouring, etc. To further ensure task diversity, we consider different patterns of hand usage. Specifically, TASTE-Rob comprises 75,389 single-hand task videos and 25,467 double-hand task videos.

Dataset	Language Instructions		Environment		Data Statistics		
	Perfect Action Alignment	Determiner	Camera	Setting	Frames	Clips	Resolution
Disney [13]	✗	✗	Dynamic	Garden	11M	8800	1080p
ADL [35]	✗	✗	Dynamic	Diverse	1M	436	960p
Charades-Ego [43]	✗	✗	Dynamic	Diverse	5.8M	7860	720p
HOI4D [29]	✓	✗	Static	Diverse	4M	2466	1080p
UT Ego [27, 30]	✗	✗	Dynamic	Diverse	0.9M	80	320p
EGTEA Gaze+ [28]	✓	✗	Dynamic	Kitchen	2.5M	10,000	480p
EPIC-KITCHENS [11]	✓	✗	Dynamic	Kitchen	6.8M	35,000	1080p
Ego4D [16]	✗	✗	Dynamic	Diverse	19.2M	83,647	1080p
<i>Ours</i>	✓	✓	Static	Diverse	9M	100,856	1080p

Table 2. **Dataset comparison with existing ego-centric HOI video datasets.** TASTE-Rob, specifically designed for generating task-oriented HOI videos, provides a diverse collection of ego-centric HOI videos with several distinctive features: static camera perspectives, precise language-action alignment, and rich object descriptors (Please see Appendix 8 for details). In the above table, we compare these key characteristics: Perfect Action Alignment (ensuring all actions in the video strictly correspond to language task descriptions), Descriptive Determiners (whether object descriptions exist), Camera Setting (fixed or dynamic camera perspective), and Environment Setting (recording locations). By the way, Even if Ego4D [16] offers an impressive 3000-hours video data, it is designed for various general tasks, and the portion applicable to HOI video generation is significantly smaller, as presented in the above table, which is only 83,647 clips. Cells highlighted in denotes the dataset with best feature in each column.

Distribution of Grasping Hands. To ensure a wide variety of hand poses, we consider two primary factors: diverse palm orientations (global pose) and varied grasping poses (detailed pose). To demonstrate the diversity of hand poses, we utilize HaMeR [34] to extract hand pose parameters and analyze the distribution based on these parameters (see further details in Appendix 9).

As shown in Table 1, we analyze the distribution of palm orientations during HOI interaction across TASTE-Rob. The analysis reveals the following insights: 1) Hand poses with palms facing down (0° - 180°) are the most common, as this orientation is practical for grasping objects. 2) Hand poses with palms facing left (90° - 270°) occur slightly more frequently than those facing right, likely because all collectors are right-handed and naturally prefer using their right hand for object manipulation.

We also provide the analysis of hand grasping pose distribution in Figures 3 and 4. Given the dominant roles of the thumb, index finger, and middle finger in HOI, we focus on examining both the inter-finger angles between these fingers and their individual curvature distributions. The analysis in Figure 3 shows a broad distribution of inter-finger angles, indicating diverse hand orientations. Figure 4 reveals two key findings: 1) The curvature distributions of the index and middle fingers exhibit similar patterns, reflecting their synchronized bending during HOI actions. 2) Our dataset captures a broad spectrum of grasping poses, driven by the variety of objects being manipulated.

3.3. Comparisons with Other Datasets

As shown in Table 2, we provide comparison between TASTE-Rob and existing ego-centric HOI video datasets.

TASTE-Rob is the first video dataset specifically designed for task-oriented HOI video generation, and it can also serve as a valuable resource for demonstrations in IL. Given that IL video demonstrations are recorded from a fixed camera perspective and include only a single action that aligns with the task instruction, we collect HOI videos under the same settings, distinguishing TASTE-Rob from other datasets. In addition, to improve comprehension of target objects, we incorporate diverse determiners of objects in language task instructions (see Appendix 8 for further details). With TASTE-Rob, we are able to generate high-quality HOI video demonstrations to achieve IL.

4. Method

Given an environment image and a task description, the generated task-oriented HOI videos are required to satisfy: 1) **Accurate Task Understanding**: correctly identifying which object to manipulate and how to manipulate it. 2) **Feasible HOI**: maintaining consistent hand grasping poses throughout the manipulation.

As illustrated in **Stage I** area of Figure 5, while videos generated by a single VDM ($\hat{v}_c$) demonstrate accurate task understanding, exhibit limited fidelity in maintaining consistent grasping poses. To fulfill both requirements, we propose a three-stage pose-refinement pipeline, outlined in Figure 5: **Stage I**: employing a learnable Image-to-Video (I2V) diffusion model to generate a coarse HOI video that meets “Accurate Task Understanding”. **Stage II**: extracting a hand pose sequence from this coarse video and refining it using a learnable Motion Diffusion Model (MDM) [45]. **Stage III**: using the refined hand pose sequence to generate a high-

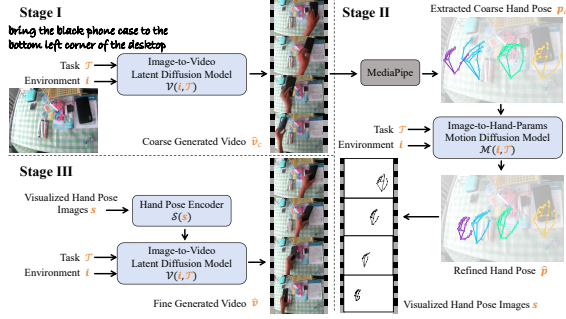


Figure 5. **Overview of our proposed three-stage pose-refinement pipeline.** **Stage I:** our learnable I2V latent diffusion model $\mathcal{V}$ generates a coarse HOI video $\hat{v}_c$. **Stage II:** extracting hand pose sequence p_c from $\hat{v}_c$, and our learnable MDM $\mathcal{M}$ refines it into $\hat{p}$. Besides, we also obtain the corresponding visualized hand pose images s of $\hat{p}$. **Stage III:** incorporating this additional pose condition into $\mathcal{V}$ to generate a fine HOI video $\hat{v}$.

fidelity HOI video that satisfies both requirements.

4.1. Preliminary: Diffusion Models

Diffusion models [17, 44] work by learning to reverse a noising process. They firstly transform data samples $x_0 \sim p_{data}(x)$ into Gaussian noise $x_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, then learn to reconstruct the original data by denoising x_T with multiple steps. With the following objective:

$$\min_{\theta} \mathbb{E}_{t, x, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \|\epsilon - \epsilon_{\theta}(x_t, t)\|_2^2, \quad (1)$$

the model $\epsilon_{\theta}(x_t, t)$ learns to predict single-step noise at each timestep t , where ϵ is the sampled ground truth noise and θ indicates the learnable network parameters. After training, the model generates $\hat{x}_0$ through a T -step denoising process from a random Gaussian noise x_T .

Video Generation. In this work, we explore HOI video generation based on DynamiCrafter [54], a powerful I2V latent diffusion model. Suppose $\mathcal{T}$, $i \in \mathbb{R}^{3 \times H \times W}$, and $v \in \mathbb{R}^{L \times 3 \times H \times W}$ denote task language descriptions, environment images, and ground truth video frames, respectively. DynamiCrafter learns the denoising process in a compact latent space: v is encoded into a compact latent space through an encoder $\mathcal{E}$, yielding latent representations $z = \mathcal{E}(v) \in \mathbb{R}^{L \times C \times h \times w}$, and decoded via the decoder $\mathcal{D}$. Within this latent space, the model performs both forward diffusion and guided denoising processes conditioned on $\mathcal{T}$ and i . Specifically, $z_0 \sim p_{data}(z)$ and $z_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The optimization objective 1 is updated as:

$$\min_{\theta_V} \mathbb{E}_{t, z, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \|\epsilon - \epsilon_{\theta}(z_t, \mathcal{T}, i, t, fps)\|_2^2, \quad (2)$$

where fps is the FPS control introduced in [54] and θ_V denotes the learnable parameters of DynamiCrafter.

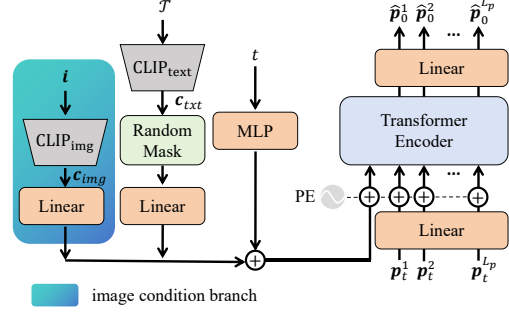


Figure 6. **overview of our learnable Image-to-Hand-Params MDM.** Based on MDM [45], we extend the framework by incorporating environment information through an additional image condition branch. The model takes a noised hand pose sequence $p_t^{1:Lp}$, t itself, a CLIP [37] text embedding c_{txt} of $\mathcal{T}$ and a CLIP image embedding c_{img} of i as input.

Motion Sequence Generation. MDM [45], utilizing a unique transformer encoder-only architecture, demonstrates remarkable performance in generating human motion sequences. Instead of predicting single-step noise, MDM directly generates clean motion sequences. Let p denotes motion sequences and $\mathcal{M}$ denotes MDM network. Here, $p_0 \sim p_{data}(p)$ and $p_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The optimization objective 1 is modified as:

$$\min_{\theta_M} \mathbb{E}_{t, p, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \|p_0 - \mathcal{M}(p_t, \mathcal{T}, t)\|_2^2, \quad (3)$$

where θ_M denotes the trainable parameters of MDM. After training, MDM generates the final clean motion sequence $\hat{p}_0$ through a T -step denoising process. However, during each step, instead of directly generating p_{t-1} through single-step denoising, MDM firstly predicts the clean motion sequence $\hat{p}_{0,t}$ from p_t with:

$$\hat{p}_{0,t} = \mathcal{M}(p_t, \mathcal{T}, t), \quad (4)$$

and then reintroduces noises to obtain p_{t-1} . Through repeating this process for T times, MDM generates the final clean motion sequence $\hat{p}_{0,0}$, denoted as $\hat{p}_0$.

4.2. Stage I: A Coarse Action Planner

Our learnable coarse action planner is designed to generate a coarse HOI video $\hat{v}_c \in \mathbb{R}^{L \times 3 \times H \times W}$ conditioned on task description $\mathcal{T}$ and environment image i . Specifically, we fine-tune DynamiCrafter [54] on TASTE-Rob to serve as the coarse action planner, denoted as $\mathcal{V}$.

Training. During fine-tuning, we adopt the training strategy similar to that of DynamiCrafter [54]. To leverage DynamiCrafter’s robust temporal capabilities while adapting it to our specific HOI video generation, we only fine-tune its image context projector and spatial layers in its denoising U-Net. The training objective remains consistent with Equation 2, with trainable parameters θ_V representing parameters of the image context projector and spatial layers.

Hand-Object Interacting Inconsistency Problem. Our generated coarse HOI videos exhibit accurate task understanding, such as identifying the object to manipulate and determining its target position. However, as shown in p_c of Figure 5, the grasping poses exhibit temporal inconsistencies during manipulation, indicating a lack of motion coherence. Specifically, these inconsistencies refer to undesirable variations in grasping poses over time, manifesting as unnatural changes in hand posture when the relative position between the hands and the grasped object should ideally remain stable. As shown in p_c of Figure 5, the **green** hand pose demonstrates a pinching gesture, which is inconsistent with the grasping posture of the **yellow** hand pose and unsuitable for the target object being manipulated.

4.3. Stage II: Revising Hand Pose Sequences

To address HOI inconsistencies, we train an Image-to-Hand-Params MDM $\mathcal{M}$. This model refines hand pose sequences $p_c \in \mathbb{R}^{L_p \times N_h \times 2}$ extracted from the coarse video $\hat{v}_c$. Specifically, we define p_c as the normalized coordinates of hand key-point sequences, where L_p denotes the length of sequence and N_h is the number of hand key-points.

Training. Our learnable $\mathcal{M}$ is designed to take the task description $\mathcal{T}$ and environment image i as input to predict refined human hand pose sequences. To achieve this, we extend the original MDM [45] framework by incorporating environmental information through an additional image branch. As shown in Figure 6, similar to the text condition branch of [45], the added image branch integrates CLIP [37] class features of environment image i . Consequently, the optimization objective 3 is updated as follows:

$$\min_{\theta_M} \mathbb{E}_{t, p, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \|p_0 - \mathcal{M}(p_t, \mathcal{T}, i, t)\|_2^2, \quad (5)$$

where θ_M denotes the parameters of $\mathcal{M}$ and ϵ denotes the sampled ground truth noise.

When we directly generate final clean pose sequence $\hat{p}$ through a T -step denoising process, $\hat{p}$ achieves physically plausible hand motions but exhibits limited spatial awareness. Conversely, p_c exhibits remarkable spatial awareness. To address this limitation, we refine p_c using $\mathcal{M}$ rather than generating from Gaussian noise. Specifically, we initialize the denoising process in Equation 4 of $\mathcal{M}$ with $p_{0, N_{rv}}$, setting p_c as $p_{0, N_{rv}}$. Through an N_{rv} -step denoising, we refine p_c to obtain the final clean hand pose sequence $\hat{p}$ which satisfies both spatial accuracy and motion feasibility.

4.4. Stage III: Regenerating with Refined Pose

With the refined hand pose sequences, we generate the fine-grained HOI videos with the additional pose condition $\hat{p}$. Inspired by ToonCrafter [55], we train a frame-independent pose encoder $\mathcal{S}$ to control the hand pose in the generated

videos. We design $\mathcal{S}$ as a frame-wise adapter that adjusts the intermediate features of each frame independently, conditioned on $\hat{p}$: $F_{inject}^i = \mathcal{S}(s^i, z^i, t)$, where s^i is the visualized image sequences of $\hat{p}$, shown as Figure 5, and F_{inject}^i is processed using a method similar to ControlNet [62].

Training. During training, we adopt a strategy similar to ToonCrafter [55], where all parameters of $\mathcal{V}$ are frozen, and only the parameters of $\mathcal{S}$, denoted as η , are trained with the following optimization objective:

$$\min_{\eta} \mathbb{E}_{t, s, \epsilon(v), \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} \|\epsilon - \epsilon_{\theta}^{\mathcal{S}}(z_t; \mathcal{T}, i, s, t, fps)\|_2^2, \quad (6)$$

where $\epsilon_{\theta}^{\mathcal{S}}$ denotes the combination of ϵ_{θ} and $\mathcal{S}$, and s represents the visualized images of hand pose sequences.

Finally, we consider our generated fine HOI videos $\hat{v}$ as the video demonstrations for IL and enable robot manipulation using policy model of Im2Flow2Act [58]. As illustrated in Figure 10, we present the imitation learning results with our generated HOI videos as demonstrations, confirming their effectiveness for enabling robot manipulation.

5. Experiments

5.1. Implementation

During training, we train our model on the training set of TASTE-Rob dataset. **Stage I:** We fine-tune our coarse action planner based on DynamiCrafter for 30K steps, with a batch size of 16 and a learning rate of 5×10^{-5} . **Stage II:** We train our MDM for 100K steps, with a batch size of 64 and a learning rate of 1×10^{-4} . **Stage III:** We fine-tune the pose encoder based on SD for 30K steps, with a batch size of 32 and a learning rate of 5×10^{-5} . During inference, we generate videos with a 50-step denoising process and refine pose sequences with N_{rv} as 10 (See more details in Appendix 12).

5.2. Evaluation

Datasets. We evaluate on two datasets: TASTE-Rob-Test and Mujoco simulation dataset. **TASTE-Rob-Test:** We split TASTE-Rob dataset into training and test sets, with the test set denoted as TASTE-Rob-Test. For fair evaluation, we select 2% of samples from each action category as test samples. **Mujoco simulation dataset:** we randomly select 50 tasks from robot simulation platform provided by Im2Flow2Act [58] for robot manipulation.

In addition, as mentioned in Section 3 and Table 2, other ego-centric video datasets feature different settings from our task, making them unsuitable as evaluation datasets.

Metrics. We evaluate the performance on three aspects: video generation, grasping pose consistency and robot manipulation. **Video Generation:** To evaluate the quality of

generated videos in both the spatial and temporal domains, we report Fréchet Video Distance (FVD) [46], Kernel Video Distance (KVD) [46] and Perceptual Input Conformity (PIC) [54] as used in DynamiCrafter [54]. **Grasping Pose Consistency:** We evaluate with three metrics - **Grasping Pose Variance** (GPV, computed by $\frac{1}{L'} \sum_{i=1}^N (\theta_i^p - \bar{\theta}^p)^2$), **Grasp Type Classification Error** (GTCE, calculating the classification error of grasping types in generated videos) and **Hand Movement Direction Accuracy** (HMDA, Measuring the directional consistency of hand movements and their alignment with the intended task trajectory), and details of these metrics are illustrated in Appendix 10. **Robot Manipulation :** We utilize videos generated on the Mujoco simulation dataset as demonstrations for IL, and deploy the policy model of Im2Flow2Act [58] for robot manipulation. The performance is evaluated by comparing the success rates.

Method	Generation Quality		
	KVD↓	FVD↓	PIC↑
consistI2V	0.24	65.77	0.85
DynamiCrafter	0.21	62.36	0.79
Open-Sora Plan	0.18	50.19	0.84
CogVideoX	0.16	48.72	0.85
TASTE-Rob	0.03	9.43	0.90

Table 3. **Quantitative comparison between TASTE-Rob and baselines.** All video generation metrics prove the better generation performance of TASTE-Rob.

5.3. Baselines and Comparisons.

We select four existing powerful I2V diffusion models - DynamiCrafter [54], consistI2V [38], Open-Sora Plan [26] and CogVideoX [60], as baselines, and conduct comparative experiments among these baselines and our method. We provide qualitative comparisons of video generation performance on TASTE-Rob-Test and wild environment in Figure 7, and quantitative comparisons on TASTE-Rob-Test in Table 3, which demonstrates superior video quality and better generalization capability of our method. According to above experiments, all existing powerful general VDMs fail to achieve manipulation tasks well, making them unsuitable for generating HOI video demonstrations. Given that the other two evaluation aspects focus on measuring the fine-grained details of generated videos, we omit further comparisons with these baseline methods.

5.4. Ablation Study

TASTE-Rob Dataset. We evaluate the performance improvements achieved by using our TASTE-Rob dataset. To be fair, we compare video generation performance between our coarse action planner (Coarse-TASTE-Rob) and Ego4D-Gen, a comparison baseline fine-tuned from Dy-

Method	Generation Quality		
	KVD↓	FVD↓	PIC↑
Ego4D-Gen	0.18	52.17	0.77
Coarse-TASTE-Rob	0.04	10.85	0.88

Table 4. **Quantitative comparison between Ego4D-Gen and Coarse-TASTE-Rob.** All video generation metrics prove the better generation performance of Coarse-TASTE-Rob.

namuCrafter [54] on 83,647 ego-centric HOI videos from Ego4D [16]. The only difference between them is the training datasets they use. As shown in Figure 8 and Table 4, our coarse action planner achieves superior performance.

Metric	Coarse-TASTE-Rob	TASTE-Rob
KVD↓	0.04	0.03
FVD↓	10.85	9.43
PIC↑	0.88	0.90
GPV↓	0.28	0.24
GTCE↓	67.8%	9.7%
HMDA↓	26.4°	11.3°
success rate↑	84%	96%

Table 5. **Quantitative comparison between TASTE-Rob and Coarse-TASTE-Rob.** All metrics prove the better generation performance, grasping pose consistency and more precise robot manipulation of TASTE-Rob.

Pose-Refinement Pipeline. We compare Coarse-TASTE-Rob and TASTE-Rob across all three evaluation aspects, as shown in Table 5. **At a coarse level**, both the video generation metrics in Table 5 and the visual comparisons in Figure 9 demonstrate that our proposed pose-refinement pipeline achieves better HOI video generation. **At a fine level**, both the grasping pose consistency metrics in Table 5 and Figure 9 further validate that our proposed pose-refinement pipeline effectively addresses the pose inconsistency issue and enables more precise robot manipulation. Besides, as shown in Figure 10 and success rate metrics in Table 5, our proposed pose-refinement pipeline helps the policy model achieve more accurate target placement, leading to improved success rates. In addition, we only do this ablation on our dataset, because videos generated by fine-tuning on existing HOI datasets (e.g., Ego4D [16]) are of low quality so that it’s difficult to extract reasonable poses for further refinement.

Denoising Steps of Pose Refinement. We also discuss the selection of N_{rv} during pose refinement. When $N_{rv} = 0$ (no refinement), hand pose sequences exhibit inconsistency issues, while $N_{rv} = 50$ (generating pose sequences from Gaussian noise) results in limited spatial awareness.

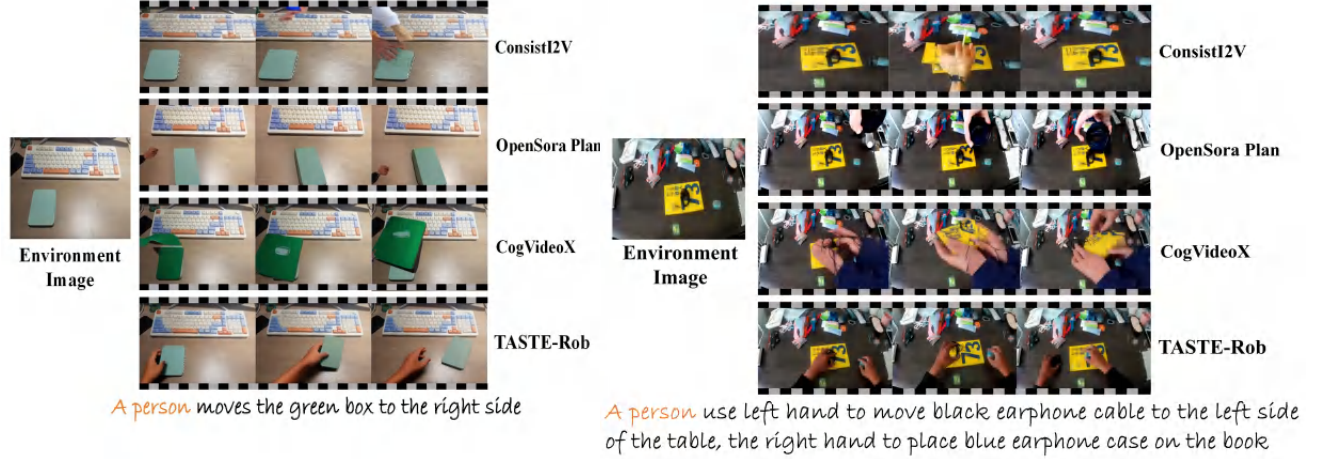


Figure 7. **Comparisons of video generation performance.** TASTE-Rob generates videos with high-quality and accurate action, while other existing powerful general VDMs fail.

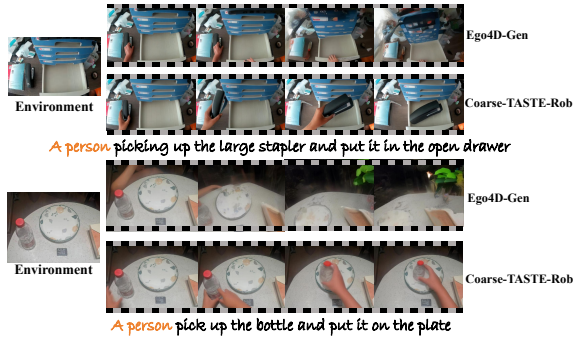


Figure 8. **Qualitative comparison between Ego4D-Gen and Coarse-TASTE-Rob.** Our TASTE-Rob dataset significantly improves the generalization of video generation models in generating HOI videos with accurate actions.

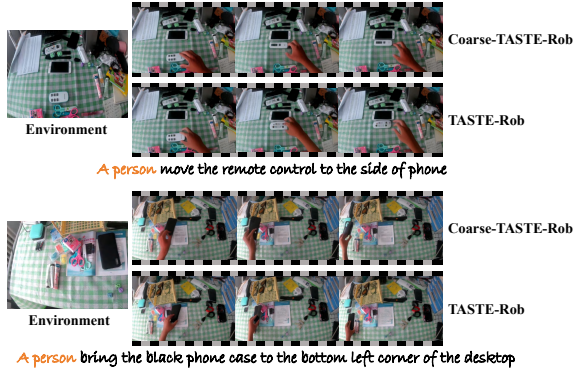


Figure 9. **Qualitative comparison between TASTE-Rob and Coarse-TASTE-Rob.** With our proposed pose-refinement pipeline, we can generate HOI videos with more accurate and feasible grasping poses.

As N_{rv} increases, we observe improved pose consistency but degraded spatial awareness. As shown in Table 6, when N_{rv} increases, video generation quality first improves and then deteriorates. Based on this trade-off, we finally select the value of N_{rv} as 10 (see more details in Appendix 11).

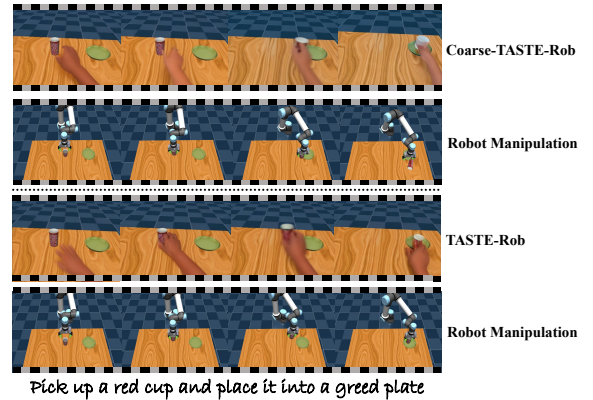


Figure 10. **Comparison of Robot Manipulation Performance.** TASTE-Rob generates higher-quality HOI videos, leading to more precise robot manipulation.

6. Discussion and Conclusion

Summary. In this work, we collect a diverse, large-scale ego-centric HOI dataset and develop a three-stage pose-refinement pipeline to generate high-quality task-oriented HOI videos. We successfully enhance HOI video generation capabilities, thereby improving robot manipulation generalization to novel scenes through imitation learning.

Limitations. Our approach achieves accurate manipulation on unseen scenes and enhances the grasping poses in generated videos. However, the generation quality degrades when objects undergo significant transformations, such as rotation and opening. In our future work, we plan to address this limitation.

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TASTE-Rob: Advancing Video Generation of Task-Oriented Hand-Object Interaction for Generalizable Robotic Manipulation

Supplementary Material

8. Details of TASTE-Rob dataset

8.1. Comparisons between Ego4D and TASTE-Rob

Ego4D [16] differs from TASTE-Rob in two key aspects: the camera perspective and the recording methodology. **Camera Perspective:** Ego4D utilized seven distinct head-mounted cameras during data collection, resulting in diverse camera perspectives throughout the dataset. In contrast, a consistent camera perspective is essential for our task. As demonstrated in Figure 12, videos in Ego4D show perspective variations in the positioning of static objects, as evidenced by the blue box in the first case. **Recording Methodology:** In Ego4D’s data collection process, participants were provided with cameras to record their daily activities. These recordings typically exceeded eight minutes in duration, and final short clips are extracted from them. Consequently, these clips often contain actions from preceding or subsequent activities, creating misalignment with the provided language instructions. This misalignment is exemplified in Figure 12 (second case, highlighted by red bounding boxes): the participant is mixing food on a plate, which contradicts the instruction “puts down the bowl”.

8.2. Design of Language Instruction in TASTE-Rob

For object manipulation tasks, it is essential to establish a one-to-one correspondence between actions and language instructions. Specifically, each language instruction should uniquely identify both the target object to be manipulated and the specific action to be performed. However, Ego4D’s [16] language instructions are insufficiently detailed to satisfy this criterion.

Identifying Unique Target Objects. The language instructions in Ego4D [16] only specify the object types without providing distinctive determiners. For instance, given the instruction “C picks up a bowl and puts down it”, when multiple bowls are present on the table, any of them could potentially be considered as the target object. In TASTE-Rob, we incorporate distinctive determiners to uniquely identify objects, such as “red bowl”, “bowl with a tomato”, and “bowl next to the kettle”. Figure 13 illustrates examples of video frames and their corresponding instructions from TASTE-Rob.

Identifying Unique Actions. Regarding object manipulation tasks, Ego4D’s [16] language instructions merely describe the action without specifying the target location. For



Figure 11. **Distribution of objects in videos from six scenes.** The diverse range of objects across different scenarios demonstrate the richness of activities captured in TASTE-Rob

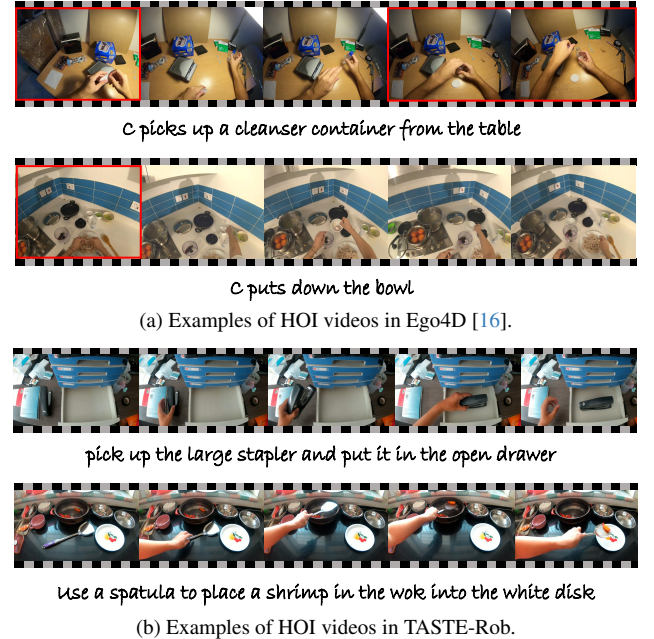


Figure 12. **Comparison between Ego4D [16] and TASTE-Rob.** Ego4D videos lack fixed camera perspective and precise action-language alignment, as shown in the red boxes, while TASTE-Rob solve these limitations.

instance, the instruction “C puts down the bowl” indicates the action “puts down” but fails to specify the intended placement location. In TASTE-Rob, we provide comprehensive instructions that include detailed spatial information, such as “take the solid black pen and put it in the pen holder” and “take the orange eraser and put it on the pencil case”.



Figure 13. **Examples of video frame and corresponding language instruction.** In TASTE-Rob, we enhance the language instructions with more specific details to ensure accurate identification of unique target objects and their corresponding actions.

8.3. Diversity of Manipulated Objects in TASTE-Rob

The distribution of manipulated objects in TASTE-Rob across different scenarios is visualized through word clouds in Figure 11. These visualizations emphasize context-dependent objects, such as kitchen utensils (bowls, spoons, plates) in cooking activities and Clothing (skirts, shorts, hats) in bed environment. Additionally, some objects appear universally across scenarios, including cups, tissues, and mobile devices. The broad spectrum of objects encountered in TASTE-Rob reflects its comprehensive coverage of diverse activities.

9. Analyzing Details of hand poses

9.1. MANO parametric hand model

Given a RGB image of hands, HaMeR [34] is able to reconstruct 3D hand meshes by predicting MANO [39] parameters (pose parameters and shape parameters). Specifically, the pose parameters are decomposed into two components: global orientation and hand pose. Global orientation parameters are represented by $\theta^g \in \mathbb{R}^{1 \times 3 \times 3}$, defining the overall 3D orientation of the hand in space. Hand pose parameters control the articulation of 15 hand joints, which are represented by $\theta^p \in \mathbb{R}^{15 \times 3 \times 3}$, defining the full hand articulation. We perform a comprehensive analysis of hand pose distribution by examining both global orientation and hand pose parameters from 16 sampled frames across TASTE-Rob videos.

9.2. Calculating Details of analyzing Distribution

Orientation Distribution. To analyze palm orientation, we first extract the normal vector of the palm plane from θ^g . This normal vector is then projected onto the frame plane (X-Y plane), and we calculate the projected angle, which tells us how much the palm’s direction deviates from the positive X-axis in the frame plane. Specifically, 0 degree points along the positive positive X-axis. We also normalize this projected angle to the 0-360 degree range. Finally, we divide the range into four 90-degree intervals to analyze the distribution of palm orientations across different directions.

Inter-finger Angles Distribution. To analyze the inter-finger angles, we calculate the angles between thumb-index and index-middle finger pairs. For each frame, we first extract the normalized direction vectors of fingers (d_t for thumb, d_i for index, and d_m for middle finger) using their joint positions. The inter-finger angles are computed as the inverse cosine of the dot product between corresponding vectors: $\arccos(d_t \cdot d_i)$ and $\arccos(d_i \cdot d_m)$. After obtaining these angles, we plot their distributions as probability density curves, and these curves illustrate how finger spreads are distributed in TASTE-Rob.

Finger curvature Distribution. To analyze finger curvature patterns, we compute the bending angle for each finger. Each finger has three joints: metacarpophalangeal (MCP) near the palm, proximal interphalangeal (PIP) in the middle, and distal interphalangeal (DIP) near the fingertip. We calculate the bending angle by obtaining two normalized direction vectors: one from MCP to PIP joint j_{M2P} , and an-

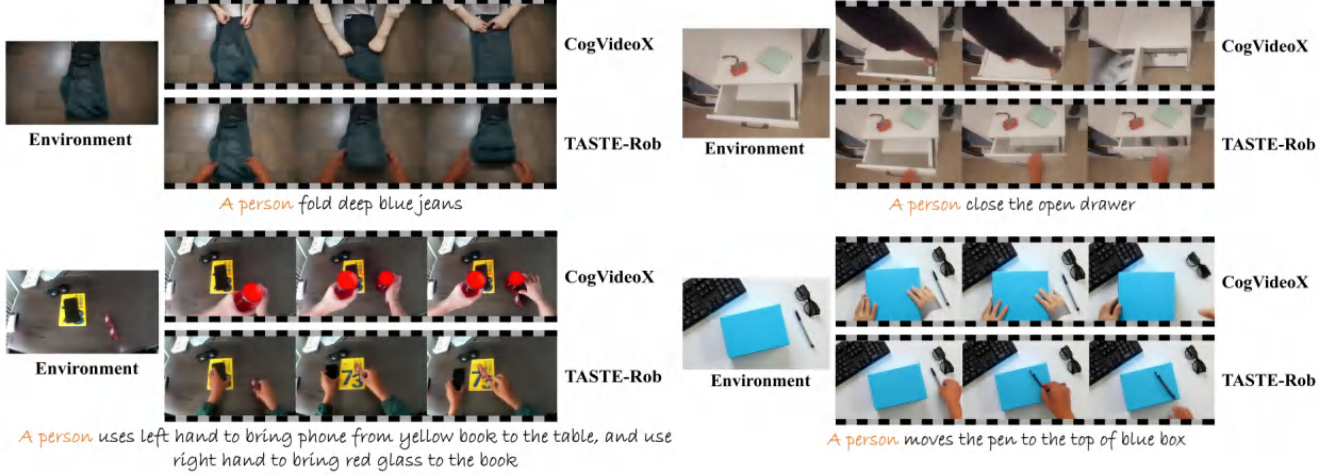


Figure 14. More results of video generation comparison between TASTE-Rob and CogVideoX [60].

other from PIP to DIP joint j_{P2D} . The bending angle is then computed as the inverse cosine of the dot product between these vectors: $\arccos(j_{M2P} \cdot j_{P2D})$. We plot these angles' distributions as probability density curves, which reveal the common curvature patterns and articulation preferences of different finger joints in TASTE-Rob.

10. Our Introduced Metrics

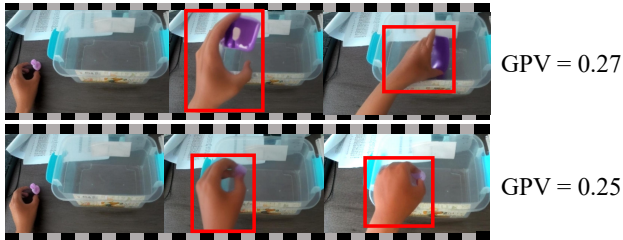


Figure 15. Examples of generated videos and corresponding GPV value. A higher GPV value indicates less consistency in grasping poses. .

Grasping Pose Variance. As illustrated in Section 9, we employ HaMeR [34] to predict accurate global orientation parameters θ^g and hand pose parameters θ^p . During grasping, variations in global orientation parameters are natural as hands need to move objects. However, hand pose parameters should remain consistent during grasping, as variations in grasping pose may lead to unstable grasps. To evaluate this consistency, we introduce Grasping Pose Variance (GPV), which computes the variance of hand pose parameters θ^p across frames containing grasping poses within a single video. Firstly, we utilize an off-the-shelf hand-object detector [40] to extract L' frames containing grasping poses from a video, where the hand-object detector can detect whether hands are interacting with objects. Then, we uti-

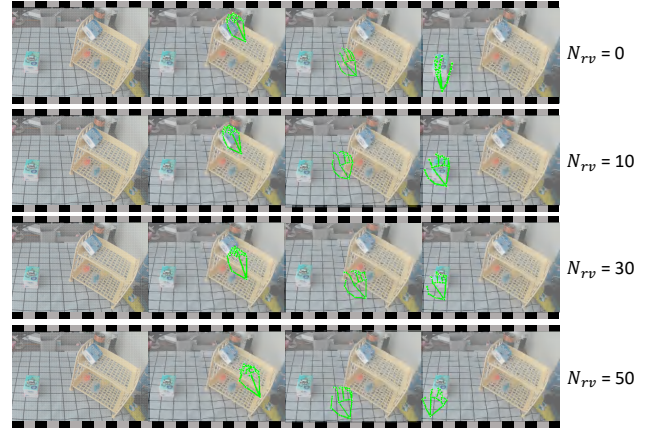


Figure 16. Visualization of different denoising steps of pose refinement. When $N_{rv} = 0$, no pose refinement is applied, whereas $N_{rv} = 50$ indicates that our learnable MDM generates pose sequences directly from Gaussian noise.

lize HaMeR [34] to predict hand pose parameters $\{\theta_i^p\}_{i=1}^{L'}$ of these frames, and calculate GPV as $\frac{1}{L'} \sum_{i=1}^N (\theta_i^p - \bar{\theta}^p)^2$, where $\bar{\theta}^p = \frac{1}{L'} \sum_{i=1}^N \theta_i^p$. Besides, for videos containing both hands, we compute GPV values for each hand independently and take their mean value.

Visualization of Grasping Pose Variance. As illustrated in Figure 15, we compare the GPV values of two generated videos. The first case displays a video generated by our coarse action planner, while the second case presents a video generated by our complete pose-refinement pipeline. The results show that the second case maintains consistent grasping poses, whereas in the first case, the grasping poses vary significantly from pinching to holding, leading to a higher GPV value. In detail, each generated video consists of $L = 16$ frames, with the corresponding grasping



Figure 17. Examples of video generation and robot manipulation in simulation platform.

sequence length L' ranging from 6 to 8 frames. For visualization clarity, we only display a few frames in Figure 15.

Grasping Type Classification Error (GTCE). First, we extract 8 frames from both the ground truth and generated videos during the manipulation process. Then, in order to detect and classify grasping types, we use HaMeR [34] to predict the corresponding pose parameters for these frames. Finally, we classify hand poses into 17 grasping types as in [14] and evaluate the generated poses by calculating the classification error against the ground truth.

Hand Movement Direction Accuracy (HMDA). HMDA assesses the quality of generated poses by quantifying the disparity in hand movement directions between generated and ground truth videos. Initially, hand keypoints are extracted from each frame of both videos via MediaPipe [31]. Subsequently, movement vectors are derived by computing the differences between consecutive keypoints. The angle differences between these vectors are then calculated, which effectively measures the divergence in hand movement directions. Ultimately, the metric value is obtained by computing the average of these angle differences.

11. More Qualitative Results

Different Denoising Steps of Pose Refinement. As illustrated in Figure 16, we refine the inconsistent grasping poses while maintaining spatial awareness through a N_{rv} -step denoising process, starting from pose sequences extracted from our coarse generated videos ($N_{rv} = 0$). During inference, we set N_{rv} as 10.

Comparisons on Video Generation. As shown in Figure 14, TASTE-Rob is able to generate high-quality HOI

N_{rv}	Generation Quality		
	KVD↓	FVD↓	PIC↑
0	0.04	10.85	0.88
10	0.03	9.43	0.90
20	0.03	9.97	0.89
30	0.04	10.04	0.90
40	0.04	10.70	0.87
50	0.06	11.27	0.88

Table 6. Quantitative comparison among different N_{rv} .

videos on various environment and tasks. We also provide corresponding videos in the supplementary materials.

Results on Robot Manipulation. As shown in Figure 17, TASTE-Rob is able to generate high-quality HOI videos on robot simulation platform and achieve accurate robot manipulation.

12. More Implementation Details

12.1. Stage I: A Coarse Action Planner

In Stage I, we fine-tune DynamiCrafter [54] on TASTE-Rob from its released checkpoint at 512 resolution. The fine-tuning process is conducted with a total batch size of 16 and a learning rate of 5×10^{-5} for 30K steps. For inference, we use a 50-step DDIM sampler and adopt the same multi-condition classifier-free guidance as in DynamiCrafter [54].

Video Sampling Strategy. Before fine-tuning, existing general I2V latent diffusion models [49, 54, 60, 63] obtain video frames v by sampling from a short clip within a longer video, with limited focus on maintaining action integrity in the generated videos. However, our HOI video generation requires strict action integrity, as it directly

impacts the effectiveness of robot manipulation learning. Therefore, we sample frames from the entire video to ensure complete action preservation. For clarity, we denote $\mathbf{v}$ as $\mathbf{v}^{0:L-1}$, and $\mathbf{v}^i \in \mathbb{R}^{3 \times H \times W}$ as the i -th frame of $\mathbf{v}$. In our work, we set $\mathbf{v}^0$ as the start frame of the entire video, $\mathbf{v}^{L-1}$ as the end frame, and obtain the intermediate frames $\{\mathbf{v}^1, \mathbf{v}^2, \dots, \mathbf{v}^{L-2}\}$ through uniform sampling across the entire video, ensuring continuity and completeness in action representation. Specifically, we set L as 16, and generate videos consisting of 16 frames.

12.2. Stage II: Revising Hand Pose Sequences

In Stage II, we train our learnable MDM [45] on TASTE-Rob. We set the 60-frame hand pose key-points normalized coordinates (extracted from videos by MediaPipe [31]) as motion parameters and train MDM to generate these coordinates conditioned on environment images and language instructions. The training is conducted with a batch size of 64 and a learning rate of 1×10^{-4} for 500K steps. During inference, MDM employs a 50-step DDIM sampler when generating from Gaussian noise. For pose refinement, we apply a 10-step denoising process from the extracted coarse hand pose sequences and adopt the same classifier-free guidance as in [45].

12.3. Stage III: Regenerating with Refined Pose

In Stage III, we fine-tune our learnable frame-wise adapter from Stable Diffusion 2. The training is conducted with a batch size of 32 and a learning rate of 5×10^{-5} for 30K steps. For training, the frame-wise adapter take the 16-frame visualized hand pose images as input. Specifically, we extract the 16-frame hand pose key-point coordinates and then draw their single-channel visualized images. For inference, we first draw visualized images from the refined hand pose sequences. Then, we generate videos conditioned on these visualized images, environment images and language instructions, applying a 50-step denoising process and the same multi-condition classifier-free guidance as ToonCrafter [55].